

Building Living Instruments for Scientific Discovery

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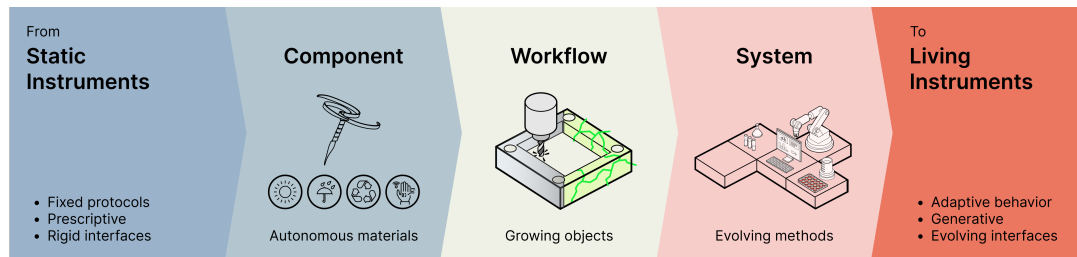


Figure 1: Instruments become extensions of scientific thinking when they can sense, grow, and evolve with our intuition.

Abstract

Experimental scientists rely on physical experimentation to validate hypotheses and make discoveries. While computational advances and AI enable unprecedented theoretical exploration, experimental instrumentation remains prescriptive, static, and difficult to modify. This gap persists due to three barriers: current “smart” instruments add complexity requiring specialized expertise, interdisciplinary work faces translation barriers between domains, and we lack accessible interfaces to connect digital design and physical execution. My dissertation develops living instruments – scientific instruments that adapt and evolve by embodying three fundamental capabilities of living systems: autonomy, growth, and evolution. I demonstrate living instruments through three approaches: materials as autonomous actuators that execute complex behaviors through biophysical programming; organisms as living fabricators where biological growth becomes manufacturing; and modular laboratory infrastructure that evolves through iteration, transforming experimental protocols into composable notebooks and replacing hardware programming with scientific scripting. Each approach removes barriers between scientific intent and physical experimentation, enabling researchers to orchestrate complex experimental workflows rather than managing technical infrastructure.

CCS Concepts

• Human-centered computing → User interface toolkits.

Keywords

Living systems design; Adaptive scientific tools; Lab automation

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1 Introduction

Digital systems have achieved unprecedented programmability. Software can be customized, scaled, updated continuously, and deployed globally. These capabilities now drive experimental science: we simulate complex multi-physics phenomena, generate millions of material designs, and optimize experiments computationally, accelerating the pace of scientific discovery [2]. Yet the physical instruments needed to validate hypotheses, verify outcomes, and execute these computationally designed experiments remain largely static and difficult to modify. This growing gap between computational sophistication and physical instrumentation raises a fundamental question: how can scientific instruments evolve to match the flexibility of our digital tools?

The gap persists due to three interconnected barriers. First, current approaches to “smart” instruments add layers of sensors and computational control, paradoxically requiring even more specialized expertise from domain scientists due to increased complexity [7]. An ecologist studying forest recovery should not need to become an electronics engineer to deploy sensor networks, yet these instruments demand exactly this. Second, interdisciplinary collaboration faces fundamental translation barriers. For instance, material scientists developing biomaterials must coordinate between biologists who understand living systems [6] and engineers who understand fabrication [15], using instruments designed for neither [7]. Third, even when tools exist and expertise is available, we lack accessible interfaces to connect them. The instruments themselves cannot bridge between digital design and physical execution; humans must serve as constant translators [12, 14, 17].

To overcome these barriers, we need instruments that can autonomously handle technical complexity, enable seamless interdisciplinary work, and create intuitive interfaces that translate intent into action [16]. For example, DNA sequencing transformed from a specialist technique to a cornerstone of modern biology when sequencers could self-calibrate, auto-correct errors, and adapt protocols independently. Similarly, 3D printing evolved from requiring extensive engineering expertise to push-button accessibility, transforming prototyping across countless fields. These breakthroughs succeeded because the instruments are programmed to handle technical complexity internally. However, most scientific instruments today across many fields lack these adaptive capabilities.

My dissertation develops **living instruments** – scientific instruments that adapt and evolve, embodying three fundamental capabilities of living systems: sensing and responding to their environment (autonomy), building structures through biological processes (growth), and improving their performance over time (evolution). These capabilities transform instruments from tools that demand expertise into systems that handle complexity independently: autonomy allows instruments to manage technical operations, growth enables instruments to harness biological processes as fabrication parameters, and evolution empowers them to continuously improve their interface between human intent and physical reality.

I realize living instruments through three complementary approaches. First, I create **autonomous instruments** that sense and respond independently through responsive materials. My self-burying seed carriers exemplify this—wood engineered to sense moisture, compute optimal planting depth through its geometric structure, and execute burial without electronics or programming [10, 11]. Ecologists can focus on restoration ecology rather than electronics engineering. Second, I develop **growing instruments** that use biological growth as a fabrication mechanism. My Fungibot platform demonstrates an open-source 3D printer that transforms mycelium growth into controlled manufacturing. The system converts fungal behaviors into material assembly processes, creating bio-integrated composites without requiring users to understand mycology [13]. Third, I build **evolving instruments** through experimental systems that improve over iteration. My modular bioreactors and automation platforms learn from each experiment, optimizing protocols and discovering new methods autonomously. Researchers can pursue ambitious questions rather than troubleshoot equipment. When instruments can sense, grow, and evolve independently, they become extensions of scientific thinking rather than barriers to it.

2 Autonomous Instruments: Embedding Sensing and Response in Materials

Large-scale reforestation exemplifies how traditional scientific instruments fail at large scale in the natural environment. Aerial seeding can quickly cover vast areas, yet germination rates remain low because of direct exposure of unburied seeds to harsh sunlight, wind, granivorous birds, as well as undesirable air humidity and temperature [3]. Conventional solutions, such as guided deployment robots, require complex infrastructure – power, electronics,

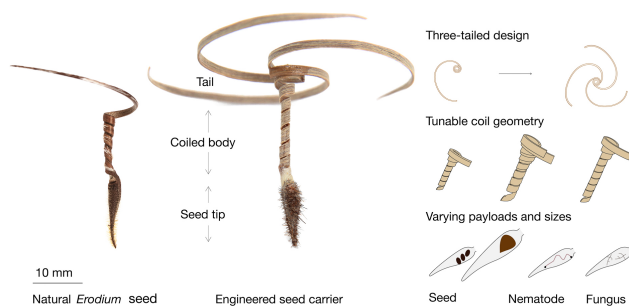


Figure 2: E-seed design and engineering. (Left) Natural *Erodium* seed versus engineered three-tailed carrier with labeled components. (Right) Parametric design variations showing tunable coil geometries and modular payloads ranging from individual seeds to fungal spores.

and perhaps even remote human operation – that limits their practical utility. What if seeds themselves could become autonomous living instruments?

Drawing from materials science knowledge of programmable materials, we developed a novel protocol for transforming wood veneer into water-responsive scientific instruments [11]. The process programs wood at the cellular level: chemical treatment enhances flexibility while maintaining structural integrity, then mechanical molding on 3D-printed forms reconstructs the cellular structure into coils with extreme curvature ($1854m^{-1}$) – 45 times larger than previously reported. The reprogrammed wood cells create a hygroscopic bilayer: upon wetting, the inner surface swells more than the outer due to fiber deformation during molding, driving the actuator to unwind. This motion reverses upon drying, creating a ratcheting mechanism powered entirely by environmental humidity cycles.

This material programming enables “more-than-nature” capabilities that transform seed carriers into precision instruments 2. While natural *Erodium* seeds fail to penetrate most soils due to insufficient anchoring from their single appendage, our three-tailed design ensures reliable deployment. Computational modeling confirmed that three tails optimize the trade-off between aerodynamic stability during aerial deployment and mechanical advantage during drilling, generating 3.7 times higher thrust than the single-tail configuration. The parameterized fabrication process creates carriers scaled to different payloads from individual seeds to fertilizer pellets. Field tests demonstrate 80% burial success across varied terrains, compared to near-zero success for natural seeds and unmodified aerial seeding (Fig. 3).

E-seed exemplifies the transformation from making smart materials in the lab to deploying materials as scientific instruments. We created a design language that translates ecological knowledge into material form through geometric parameters (Fig. 2). The barrier between domain expertise and technical implementation dissolves as the material itself becomes the experimental apparatus, operating through fundamental physical properties rather than external control.



Figure 3: E-seed as self-operating instrument. The hygroscopic wood veneer converts moisture cycles into mechanical rotation, executing a complete experimental protocol autonomously. Each phase, from aerial deployment to germination at ample humidity, emerges from material-environment interaction.

3 Growing Instruments: Harnessing Biological Growth for Fabrication

Traditional biofabrication approaches treat living organisms as factories: input nutrients, output products. Engineered living materials (ELMs) represent a new class of materials formed or assembled by living organisms. They can be responsive, self-repairing, or act as manufacturing substrates [4]. Fungi exemplify this potential: mycelium, the root-like network of fungi, not only produces material but senses, adapts, and optimizes growth patterns based on environmental conditions. This living network autonomously strengthens weak points, creates water-tight seals without adhesives, and adapts material properties based on local stresses. However, accessing these capabilities requires specialized knowledge of sterile culture techniques, controlled growth environments, and biological timescales that limit experimentation to specialized laboratories [6]. What if we could transform living organisms into accessible experimental instruments?

My work on mycelium biocomposites demonstrates how to create growing instruments that harness biological processes [13].

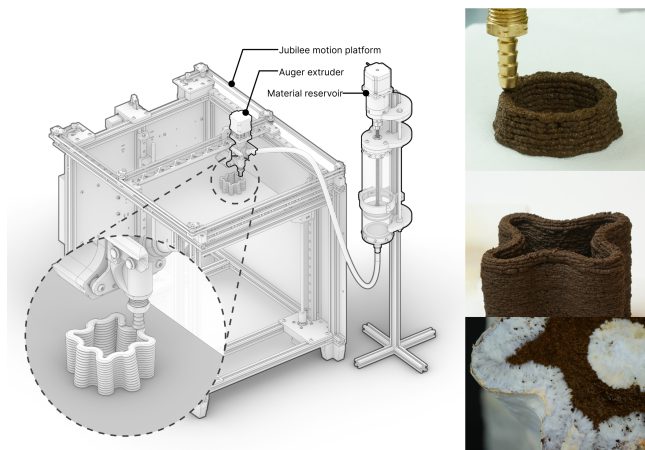


Figure 4: Mycelium as method: Fungibot enables direct experimentation with living materials. The organism simultaneously serves as manufacturing system, experimental subject, and active collaborator in developing new biofabrication protocols.



Figure 5: Bio-welding demonstration with 3D-printed Moai statue. Two halves printed flat, colonized for one week until mycelium covered outer surfaces, then pressed together for second incubation. Mycelium growth across the seam fuses the halves into a single homogeneous piece.

Through Fungibot, an open-source 3D printing platform, we enable researchers to experiment with living materials as easily as they work with conventional 3D printing. Realizing this vision required addressing a three-fold challenge across materials science, mycology, and engineering. First, we convert biowaste into fabrication materials. Our formulation of printable paste, Mycofluid, achieves 73% coffee ground content (the highest reported in literature) while maintaining rheological properties for 3D printing. Second, we develop 3D printing workflows compatible with mycelium incubation for rapid prototyping and form experimentation. Our growth protocol works in non-sterile environments, eliminating cleanroom requirements. Third, we create Fungibot that bridges digital design and biological growth by parameterizing both printing variables (layer height, infill density, print speed) and incubation conditions (humidity, temperature, growth duration).

Fungibot instrumentalizes growth as a fabrication parameter to create new material capabilities. During incubation, mycelium transforms discrete printed layers into continuous biocomposites through hyphal penetration, increasing toughness and impact resistance of the final prints. Beyond reinforcement, we harness mycelium's bio-welding capability: separately printed components placed in contact during incubation fuse into unified assemblies without adhesives or fasteners (Fig. 5). Material properties emerge from the controlled interplay of printing parameters and growth conditions.

This work presents an instrument that transforms biological growth from an uncontrolled variable into a controlled fabrication workflow. By bridging digital design and biological behavior, Fungibot demonstrates how experimental tools can grow alongside research progress, enabling designers, architects, and scientists to harness living materials without becoming mycologists.

4 Evolving Instruments: Experimental Systems that Learn

The disconnect between computational hypothesis generation and physical validation creates a fundamental bottleneck in scientific discovery. While artificial intelligence (AI) can propose millions of material compositions or reaction conditions, testing these possibilities requires manual laboratory work that has not changed significantly in decades. Current laboratory automation remains in well-funded institutions with dedicated engineering teams and locks researchers into rigid protocols. This gap in experimental reality limits who can contribute to scientific discovery and how

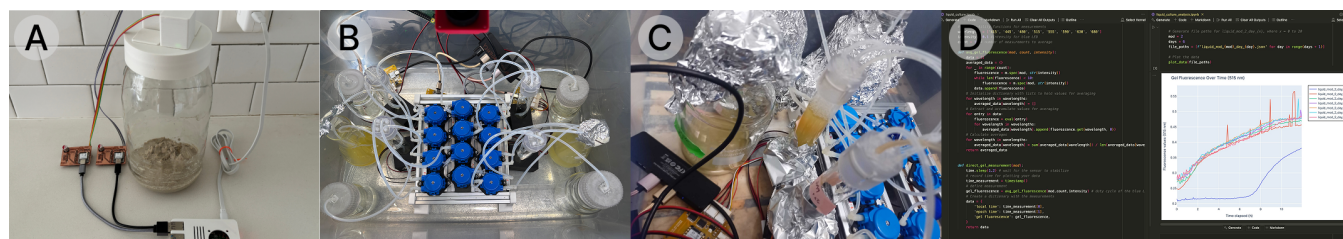


Figure 6: Bioreactor is a modular toolkit for experimental synthetic biology that can be configured to execute experimental protocols from monitoring fermentation progress (A) to parallel experimentation with genetically modified yeast with four parallel sets (B) to frequent sampling of mechanistically undefined liquid culture (C). Hardware communication, data analysis, and visualization can be done in the notebook environment (D).

quickly we can validate new ideas. What if experimental workflows could adapt and optimize like living systems?

Synthetic biology is transforming manufacturing by engineering organisms to produce everything from medicines to materials. Central to this field is the ability to design genetic circuits computationally, then test whether cells behave as predicted. However, validating these designs requires measuring gene expression (typically by tracking fluorescent proteins) through labor-intensive sampling every few hours around the clock. Missing a critical time point can mean missing the mechanistic insight that explains why a design succeeded or failed.

In this project, our modular bioreactor system demonstrates how living workflows transform this process. Based on the special experimental requirements of synthetic biology research and common laboratory practices, we decompose experimental workflows into modular components, including networked microcontrollers for sample handling and data collection, plug-and-play sensors, and software in the format of computational notebooks that integrate with existing data processing units and orchestrate hardware operations. The modular footprint of hardware and software gives researchers the flexibility to configure experimental systems as they need, scaling experimental workflow up without adding complexity to programming (Fig. 6). In practice, the system continuously tracks bio-signals from the reactor, capturing reaction dynamics at finer temporal resolutions that is otherwise impractical through manual sampling. This fine-grained data revealed unexpected oscillations in our test, patterns that would be omitted if the reaction was sampled at normal work hours. Without automation, these patterns would require a team working in shifts; with our system, a single researcher captures weeks of high-resolution data. More importantly, the platform integrates physical instrumentation with analytical programs that are readily familiar to synthetic biologists. Researchers write experimental protocols in a notebook environment, where one cell might start overnight cultures, another monitors growth rates, and a third applies signal processing to identify reaction patterns. When interesting behaviors emerge, the same notebook can learn and modify the experimental protocol or adjust environmental conditions. This tight loop between observation and experimentation accelerates the iterative design process essential to synthetic biology.

By transforming laboratory practices into computational exploration, the system demonstrates how experimental workflows can

become living systems that evolve with science. Graduate students can build custom bioreactors for specific organisms [5]. Community biolabs can create screening platforms for citizen science projects [8]. Educators can teach experimental design through hands-on automation [9]. This establishes programmable infrastructure where workflows are living entities that adapt, learn, and improve. Rather than fixed automation that constrains scientists to follow predetermined protocols, these systems evolve with research questions. The experimental protocol becomes as dynamic as the biological systems it studies, creating a future where laboratories themselves exhibit the adaptive intelligence we seek to understand in living organisms.

5 Future Work

The living instruments developed in this dissertation establish foundations for increasingly sophisticated forms of material intelligence and experimental automation. Looking forward, several critical directions emerge that will shape how we design, discover, and deploy intelligent scientific instruments.

Self-driving laboratories represent the natural evolution of modular automation, where experimental systems not only execute protocols but also intelligently design and adapt them [9]. Future work will explore how to integrate machine learning models directly into experimental workflows, enabling bioreactors that optimize growth conditions in real-time, materials that learn from environmental feedback, and fabrication systems that iteratively improve their processes. This requires developing new interfaces between physical instrumentation and AI systems, standardizing data formats for experimental results, and creating feedback loops that translate observations into actionable parameter adjustments.

The role of human creativity in automated discovery presents fundamental questions about agency and expertise. To answer these questions, I organized and participated in the Pathways to Open-Source Hardware workshops [1], where scientists come together to discuss and practice new ways to automated discovery using open-source hardware. It is only the beginning to realize that the tension between standardization (needed for interoperability) and novelty (needed for innovation) requires careful navigation. Future systems must amplify rather than replace human intuition. As HCI researchers, we need to work with domain experts closely to develop a common language that bridges computational insights and tacit domain knowledge.

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